

Introduction

As part of my cybersecurity undergraduate learning journey, I have explored tools and technology that modern security and infrastructure teams work by to automate, secure, and scale their environments. In this volume I studied Ansible, an infrastructure-as-code automation platform widely used across the industry. My interest is to manage, configure, deploy services, and learn how to reduce operation risk. As I develop my interest in how automation works with security consulting roles I realize how closely these concepts align with my skills required for consulting roles. This report outlines the core components of Ansible, the YAML language that powers it, and I applied these concepts by deploying my own Docker-based server using scripted playbooks.

Understanding the Language Behind Ansible: YAML

Learning multiple programming languages throughout my cyber program has taught me the importance of readable consistency when working with complex infrastructures. Furthermore, Ansible supports *YAML (Yet Another Markup Language)*, a human-readable data serialization language used for configuration files and data exchange between languages with different data structures” (GeeksforGeeks, 2024). YAML's simplicity reduces the likelihood for misconfigurations, which is critical in environments where analysts must manage many systems efficiently and securely.

In my project, YAML served as the foundation for my Docker Compose configuration. This allowed me to define containers, services, and environment variables in a structured, repeatable format. The same YAML file could be reused across environments, demonstrating one of Ansible’s core strengths for an analyst: repeatability without complexity.

Core Ansible Concepts and Why They Matter

To understand how Ansible automates IT tasks, it's important to break down its building blocks.

These features are ideal for big servers with distributed applications. Which is exactly the type of environments Cisco consultants support:

<u>Modules</u>	Each module performs a single, specific task(e.g_ creating a cloud instance, starting containers). Modules are granular and reusable.
<u>Playbooks</u>	A playbook is a sequence of tasks written in YAML. It defines what to execute, where to execute it, and which user should run the tasks.
<u>Hosts/Inventory</u>	A file that groups IP addresses/hostnames, letting analysts target multiple machines at once.
<u>Agentless Architecture</u>	Ansible only requires SSH access to target machines. No agents need to be installed, reducing overhead and security risk.

Applying Ansible to Docker: My Project

To deepen my understanding, I deployed my own server using Docker and automated the process with Ansible. The workflow included:

Writing a docker file to define containers configurations, environment variables, log directories, and startup scripts
Using Docker Compose (YAML) to orchestrate the container and its services.
Creating an Ansible playbook to automate container deployment, configuration and environment setup

Ensuring the same playbook could reproduce the application across multiple environments

To continue, this hands-on experience demonstrated how Ansible integrates with containerized environments, this is an important skill as companies have migrated to cloud-architectures.

My project demonstrates not only my technical understanding but also the mindset of repeatability, scalability, and security, which are qualities for a Cisco engineer.

Conclusion

Through studying Ansible and applying it to a Docker-based deployment, I gained practical experience with automation, configuration management, and infrastructure as code. These skills are directly relevant to the responsibilities of a Cisco Security Consulting Engineer, who must design secure, scalable and repeatable solutions for clients. By learning how modules,playbooks, and YAML-based configuration work together, I developed a strong foundation in automating IT tasks and managing cloud environments. This project not only strengthened my technical abilities but also reinforced my interest in pursuing automation-driven security engineering at Cisco.